Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Participatory Brain Rehabilitation Training Solution Using Acoustic Stimulation			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio • Medical		Bio • Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Functional Restoration / Assistive & Welfare Devices	Cognitive / Sensory Function Support Devices	Therapeutic Devices & Diagnostic Devices	Other Therapeutic and Diagnostic Devices
Concept of Proposed Technology	The proposed technology is a **participatory digital therapeutic for brain rehabilitation** that utilizes **acoustic stimulation and voice-based interaction** to address the limitations of conventional cognitive rehabilitation methods. Traditional brain rehabilitation typically relies on repetitive training paradigms that often result in boredom, reduced motivation, and low long-term adherence. These limitations hinder sustained therapeutic effectiveness. To overcome these challenges, the proposed solution integrates **AI-driven hyper-personalization, gamification, and real-time interactive responsiveness** to deliver an engaging and adaptive rehabilitation experience. The system is designed to actively involve users in training tasks while dynamically adjusting content based on individual performance and response patterns.			

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	The proposed solution is intended for **brain rehabilitation and cognitive function enhancement**. It can be deployed as: • A complementary tool alongside conventional rehabilitation therapies, or • A standalone digital cognitive training solution
Rationale for Joint R&D Project	The objective of this project is to develop an **AI-based digital therapeutic solution suitable for global deployment**. Achieving this goal requires incorporation of **language- and culture-specific acoustic and vocal characteristics**, which cannot be sufficiently addressed through single-country development. Joint R&D with overseas research institutions enables: • Acquisition of **multilingual and cross-cultural voice datasets** • Enhancement of AI engine robustness across diverse linguistic environments • Development of culturally adaptive content and real-time feedback mechanisms • Validation of the solution in local, real-world user settings This international collaboration is essential to verify global applicability, improve localization competitiveness, and establish commercialization readiness across multi-country markets.
Key Words	1. AI-Based Digital Therapeutics 2. Hyper-Personalized Cognitive Rehabilitation 3. Acoustic Stimulation and Voice-Based Interaction 4. Real-Time Interactive Feedback System 5. Multilingual and Cross-Cultural Localization Technology

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	• Development of AI-based acoustic and voice-centered digital therapeutics technologies for global use • Development of AI analysis engines utilizing multilingual and cross-cultural voice data • Advancement of hyper-personalized cognitive rehabilitation training algorithms • Implementation of real-time interactive feedback technologies • Development of AI models balancing therapeutic effectiveness and user experience • Establishment of a scalable digital therapeutics platform architecture			
R&D Target Specifications	**Multimodal Data Collection** • Collection of at least **300 multimodal datasets**, including voice, acoustic responses, and training performance logs **AI Engine Performance for Screening** • Development of AI models for screening **hearing loss and mild cognitive impairment (MCI)** • Achievement of **F1-score, sensitivity, and specificity ≥ 0.85** **Auditory Cognitive Rehabilitation Content** • Development of at least **five (5) auditory cognitive rehabilitation modules** targeting attention, memory, reaction speed, and communication-related cognitive functions **Pilot Validation and User Testing** • Pilot studies conducted across clinical, rehabilitation, and general user environments • Recruitment of at least **100 participants** for validation and usability evaluation **User Interface and Usability** • Age-inclusive UI/UX design across user groups • User satisfaction scores of **≥ 85%** in usability assessments			
R&D Plan	The project will be executed through a phased approach encompassing data acquisition, AI engine development, therapeutic content creation, pilot validation, and platform scalability enhancement. International collaboration will support multilingual data collection and real-world validation. **Key Tasks and Methodologies** **(1) Multimodal Data Collection and Preprocessing** • Collection of multilingual and cross-cultural voice and acoustic data through overseas collaboration • Dataset construction (≥ 300 samples) with quality verification • Data preprocessing, labeling, anonymization, and dataset preparation **(2) AI Engine Development for Screening and Personalization** • Development of AI models for hearing loss and MCI screening using acoustic features			

- Enhancement of hyper-personalized training algorithms
- Internal validation targeting performance metrics ≥ 0.85

(3) Auditory Cognitive Rehabilitation Content Development

- Design and implementation of auditory rehabilitation modules
- Integration of AI-driven adaptive difficulty control

(4) Real-Time Interactive Feedback and UI/UX Design

- Implementation of real-time visual and auditory feedback
- Development of age-inclusive UI/UX
- Iterative usability testing and refinement

(5) Pilot Validation and Performance Evaluation

- Pilot testing in hospitals, rehabilitation centers, and home environments
- Recruitment of ≥ 100 participants
- Evaluation of AI performance, therapeutic effectiveness, and usability

(6) Scalable Platform Architecture Establishment

- Design of a modular and extensible platform architecture
- Enhancement for future content expansion and global deployment

Development Timeline

- **Year 1:** Data collection, preprocessing, and foundational AI model development
- **Year 2:** AI engine advancement, therapeutic content development, and UI/UX implementation
- **Year 3:** Pilot validation, performance evaluation, and platform scalability enhancement

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will serve as a **core technology and research partner** for this project, focusing on the development, validation, and global applicability of AI-based acoustic and voice analysis technologies. • **Data Acquisition & Management**: Collect, preprocess, and perform quality control on multilingual and cross-cultural voice datasets to ensure robustness and generalizability. • **Experimental Validation**: Participate in experimental validation processes, including pilot studies, to verify technical feasibility and clinical relevance. • **Scientific & Methodological Advisory**: Provide scientific, analytical, and methodological advisory support throughout the project lifecycle. • **Joint R&D & Dissemination**: Actively collaborate in joint R&D activities and contribute to dissemination of research outcomes through publications and academic presentations.
Expected R&R of Korean SME Partner	**(1) Technology Development and Implementation** • Design and development of AI-based acoustic, speech, and cognitive analysis technologies • System integration and optimization for service deployment **(2) Platform and Content Development** • Development and operation of the digital healthcare platform • Design and implementation of user-centered UI/UX • Development of cognitive and auditory digital content **(3) Testing, Validation, and Demonstration** • Execution of real-world testing and demonstration • Evaluation and improvement of system stability and usability **(4) Quality Management and Documentation** • Establishment of quality management processes • Preparation and management of technical documentation **(5) Collaboration and Commercialization Support** • Collaboration with research partners • Support for service application and commercialization activities

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	Technological Outcomes • Establishment of a **globally applicable AI-based digital therapeutics core technology** centered on acoustic and voice analysis • Validation of hyper-personalized, interactive rehabilitation mechanisms in real-world environments • Development of a scalable and modular digital therapeutics framework
	Commercialization Potential • Readiness for deployment in hospitals, rehabilitation centers, and home-based care environments • Applicability as both standalone and complementary rehabilitation solutions • Expansion potential through partnerships with healthcare and digital health stakeholders
	Industrial and Economic Impact • Strengthening of international collaboration in AI-based digital healthcare • Acceleration of data-driven digital therapeutics innovation • Contribution to economic value creation through global healthcare services expansion

6. References

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